Resolution Geometry

Presentation and Thresholds: What Declarations Can and Cannot Change

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Abstract

Two objections claim that Resolution Geometry (RG) is controlled by human setup.

- *Presentation.* The effective symmetry group depends on the latent presentation, so retyping a presentation without changing any data could flip LOCKED statuses.
- *Thresholds.* Since exact law equality cannot be certified from finite samples, confidence thresholds (α, τ) enter, and statuses become arbitrary functions of hand-typed numbers.

Both objections contain a correct premise and an incorrect conclusion. The liftable symmetry is indeed presentation-relative, but the observable layer and the *observable symmetry* Γ_{obs} are not. Every result whose proof uses the symmetry only through its action on the quotient holds with Γ_{obs}, so no LOCKED observable result can flip under a change of presentation. Thresholds do enter, but provably unavoidably (Le Cam), and in a controlled way: statuses form a monotone filtration in stringency, each certificate can be reported as a threshold-free *critical margin*, and VACUOUS and OUTSIDE DOMAIN never depend on thresholds.

One genuine improvement results. The print edition now states $\Gamma \leq \Gamma_{\mathrm{obs}}$ and places the forcing rungs on Γ_{obs}. A seven-check battery passes 7/7 on five seeds, and two canaries are caught.

1 Verdicts

claim	status	reason
Γ is presentation-relative	correct (already in L6)	liftability depends on fibre structure
retyping the presentation flips LOCKED observable results	false	observable certificates are functions of the law image and the data (DF01, DF03)
symmetry-forced results are presentation-relative	false for forcing via the action on Q	they hold with Γ_{obs} (DF02); print repaired
thresholds are arbitrary human inputs	they are declarations, and unavoidable	Le Cam minimax bound (DF06)
statuses are unstable in (α, τ)	false	monotone filtration (DF04); critical margins (DF05)
LOCKED versus VACUOUS is decided by α-gates	false	VACUOUS depends only on the declared task (DF07)

2 Presentation

Theorem 2.1 (Observable invariance)**.** *Fix the protocols, the relabelings, the task and the declarations. Then:*

- *the quotient* $Q \cong \operatorname{im} F$*;*

- *every natural module, once its law-space structure is fixed;*
- *task descent;*
- *every certificate computed from data and declarations*

depend on the latent presentation only through $\operatorname{im} F$.

Proof. Each is a function of law values (and of data drawn from those laws). Two presentations with the same law image and the same declarations give the same inputs to each construction. □

Definition 2.2. The *observable symmetry* is $\Gamma_{\mathrm{obs}} = \{\psi \in \Psi : \psi(\operatorname{im} F) = \operatorname{im} F\}$. The *liftable symmetry* Γ consists of those elements that come from Stab.

Proposition 2.3 (Forcing uses the observable symmetry)**.** $\Gamma \leq \Gamma_{\mathrm{obs}}$ *as actions on* Q*, and* Γ_{obs} *is presentation-invariant. Statements whose proofs use symmetry only through its action on* Q *hold with* Γ_{obs}*:*

- *invariant forms and isotropy forcing, including the light cone;*
- *metric proportionality;*
- *the anchor budget.*

Statements that use lifts remain presentation-relative and are typed as such: non-splitting, gauge arguments and latent transport.

Measured on the Gaussian location family in two presentations (DF02):

- the liftable translations number 41 and 1 of the 41 tested;
- Γ_{obs} contains all 41 in both presentations;
- the invariant forms of the observable isotropy are 1-dimensional in both.

On 300 random pairs of finite presentations with different fibre sizes, the quotient, the Hellinger distances and the law-determined tasks are identical (DF01).

DF03 checks the dispatcher. Its law-defined requests receive only the law image and the data, so their statuses cannot depend on the presentation (0 flips in 240). Latent-indexed requests, such as lifts and gauge splittings, can change: 82 of 160 did. This is a signature check, verifying that the presentation is not an input of law-defined certificates. Its canary, which leaks the presentation into those certificates, is caught.

Remark 2.4 (What the critic can do). A critic who changes the presentation can change three things:

- the liftable symmetry;
- the gauge structure;
- every latent claim.

RG declares these presentation-relative (L0, L6). The critic cannot change any statement about the observable experiment.

3 Thresholds

Theorem 3.1 (Thresholds are unavoidable; [2])**.** *For two laws whose n-sample distributions are at total-variation distance t, every decision rule has worst-case error at least $(1-t)/2$.*

With $n = 100$ Bernoulli samples, the bound is 0.16, 0.46 and 0.496 for $\delta = 0.1$, 0.01 and 0.001 (DF06). A finite-data method therefore either declares an error budget or hides one. RG declares it: L48, L56, and the preregistration requirement H-adm.

Proposition 3.2 (Monotone filtration and critical margin)**.**

- *For fixed data, the set of LOCKED requests is non-increasing in stringency (smaller α, larger τ).*
- *The* critical margin*, the largest τ at which a certificate locks, is a statistic of the data alone. For a rank certificate it is a lower confidence bound that converges to the true value.*

Measured:

- over 200 instances and 20 threshold pairs, no stricter gate locked where a looser one did not (DF04);
- a canary that adds threshold-dependent noise is caught;
- for a true value of 0.08, the margins are 0.003, 0.033, 0.053 and 0.065 at $n = 500$, 2000, 8000 and 32000 (DF05).

Reporting margins turns "LOCKED at $\tau = 0.05$" into "locks up to $\tau = 0.065$". The reader may then apply any declared threshold.

Remark 3.3 (VACUOUS and OUTSIDE DOMAIN). These statuses depend only on the declared task and domain (DF07, a signature check of the dispatcher order). Thresholds act only among LOCKED, ESCALATE and STRUCTURAL WALL[τ], and the last carries its τ in its name.

4 What declarations decide

output	depends on	invariant under
quotient, observable modules, tasks	law image (and fixed law-space structures)	presentation
observable symmetry, forcing on Q	law image, Ψ	presentation
liftable symmetry, gauge, latent claims	presentation (declared)	reparametrization of the same presentation
certificate status	data, thresholds (monotone)	presentation
critical margin	data	thresholds, presentation
VACUOUS, OUTSIDE DOMAIN	declared task, domain	thresholds, presentation

Declarations decide which questions are asked, and how strictly they are certified. They do not decide the answers to observable questions, and they cannot scramble the order of certified claims.

A Reproducibility

`rg_defense_attack.py` passes 7/7 on seeds 20261023, 1, 42, 777 and 2026. Its canaries are a presentation leak into law-defined certificates and threshold-dependent noise; both are caught. DF03 and DF07 are signature checks of the dispatcher; the substantive checks are DF01, DF02 and DF04–DF06.

References

[1] M. Rada, *Resolution Geometry* (print edition), 2026.

[2] L. Le Cam, *Asymptotic Methods in Statistical Decision Theory*, Springer, 1986.